entropy for discrete random variables

1. Shannon entropy $H(p)$

1.2 $H(p) := -\sum_{k=1}^{K} p(X=k)\log p(X=k)$

$= -\sum_{x\in X} p(x)\log p(x)$, $|X|<\infty$

1.2.2 $H(p)\geq 0$

1.3 $H(X) := H(p)$, if $X\sim p$, p is distribution

1.4 binary entropy function $H(\theta)$

$H(\theta) := H(p) = -\{p(X=1)\log p(X=1) + p(X=0)\log p(X=0)\}$

$= -[\theta\log\theta + (1-\theta)\log(1-\theta)]$

here $\theta = p(X=1)$

2. conditional entropy

2.2 $H(X|Y) := \sum_y p(y) H(P_{X|Y=y})$

$= \sum_y p(y) H(X|Y=y)$

$= \sum_y p(y)\left(-\sum_x p(x|y)\log p(x|y)\right)$

$= -\sum_{x,y} p(x,y)\log p(x|y)$

here $P_{X|Y=y}$ is the distribution of X conditional on $Y=y$

3. cross entropy $H(p,q)$ (for discrete distributions)

3.2 $H(p,q) := -\sum_k p_k \log q_k$

$= -\sum_{x\in X} p(x)\log q(x)$

3.3 $H(p) = H(p,p)$

3.4 binary cross entropy $H(\theta_p,\theta_q)$

3.4.2 $H(\theta_p,\theta_q) := H(p,q) = -\sum_{x\in X} p(x)\log q(x)$

$= -\{p(X=1)\log q(X=1) + p(X=0)\log q(X=0)\}$

$= -[\theta_p\log\theta_q + (1-\theta_p)\log(1-\theta_q)]$

here $\theta_p = p(X=1)$, $\theta_q = q(X=1)$

3.4.3 If $\theta_p\in\{0,1\} \Rightarrow H(\theta_p,\theta_q) = -\log(\theta_p\theta_q - (1-\theta_p)(1-\theta_q))$

proof: $H(\theta_p,\theta_q) = \begin{cases} -\log\theta_q, & \text{if } \theta_p = 1 \\ -\log(1-\theta_q), & \text{if } \theta_p = 0 \end{cases}$

$=$ right ∎

4. KL divergence (relative entropy) (for discrete distributions)

4.2.1 $KL(p\|q) := \sum_k p_k \log\frac{p_k}{q_k}$

$= \sum_{x\in X} p(x)\log\frac{p(x)}{q(x)}$, $\forall p,q\geq 0$.

here $0\log\frac{0}{q(x)} := 0$ if $q(x)\geq 0$

$p(x)\log\frac{p(x)}{0} := +\infty$ if $p(x)>0$

4.2.2

$KL(p\|q) := \sum_{x\in A} p(x)\log\frac{p(x)}{q(x)}$, $\forall p,q\geq 0$

here $p(x)\log\frac{p(x)}{0} := +\infty$, $\forall x\in A$.

$A = \{x\in X \mid p(x)>0\}$, support of p.

4.3 $KL(p\|q) = \sum_x p(x)\log p(x) - \sum_x p(x)\log q(x)$

$= H(p,q) - H(p)$

4.4 $f(x) = x\log\frac{x}{c}$, $x>0, c>0$

$\Rightarrow \lim_{x\to 0^+} f(x) = 0$

$\because KL(p||q) \geq 0$

② If $KL(p||q)=0$

$$\Rightarrow \begin{cases} \sum_{x\in A} p(x)\log\frac{q(x)}{p(x)} = \log\sum_{x\in A} p(x)\frac{q(x)}{p(x)} \\ \sum_{x\in A} q(x) = \sum_{x\in X} q(x) \end{cases}$$

$$\Rightarrow \begin{cases} \frac{q(x)}{p(x)} = c, \forall x\in A, \text{ c is some constant.} \\ \sum_{x\in A} q(x) = 1 \end{cases}$$ (see 02232.6)

$$\Rightarrow c\sum_{x\in A} p(x) = c\sum_{x\in X} p(x) = c = 1$$

$$\Rightarrow q(x) = p(x), \forall x\in A$$

$$\Rightarrow q(x) = p(x), \forall x\in X$$

③ $p(x)=q(x), \forall x\in X$

$\Rightarrow KL(p||q)=0$

▨

5. Jensen-Shannon divergence

5.2 $JS(p_1,p_2) := \frac{1}{2}(KL(p_1||q)+KL(p_2||q))$, $q=\frac{1}{2}(p_1+p_2)$

5.3 $JS(p_1,p_2) = JS(p_2,p_1)$

$\because 0\log\frac{0}{q(x)} := 0$ for $q(x)\geq 0$

$\therefore KL(p||q) = \sum_{x\in X} p(x)\log\frac{p(x)}{q(x)} < \infty$

4.6 If $|X|<\infty$, then,

$KL(p||q)\geq 0$,

$KL(p||q)=0 \Leftrightarrow p=q$

proof: If $\exists x_0$, s.t., $p(x_0)>0$, $q(x_0)=0$

$\Rightarrow KL(p||q) = +\infty \geq 0$ (by 4.5.①)

$\therefore$ next, we let $support(p)\subseteq support(q)$,

i.e., we let $KL(p||q)<\infty$

let $A=\{x\,|\,p(x)>0\}$, support of $p(x)$

① $-KL(p||q) = -\sum_{x\in X} p(x)\log\frac{p(x)}{q(x)}$

$= -\sum_{x\in A} p(x)\log\frac{p(x)}{q(x)}$

$= \sum_{x\in A} p(x)\log\frac{q(x)}{p(x)}$ (here $\frac{q(x)}{p(x)}>0$)

$\leq \log\sum_{x\in A} p(x)\frac{q(x)}{p(x)}$ ($\overset{\text{see 02232}}{\text{Jensen's inequality}}$)

$= \log\sum_{x\in A} q(x) \leq \log\sum_{x\in X} q(x) = \log 1 = 0$

proof: $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x\log_b x - \lim_{x\to 0^+} x\log_b c$

$= \lim_{x\to 0^+} x\log_b x$

$= \lim_{x\to 0^+} \frac{x\ln x}{\ln b}$

$= 0$ (see 02233)

4.5 If $|X|<\infty$, then

$KL(p||q)<\infty \Leftrightarrow support(p)\subseteq support(q)$

proof: ① $\Rightarrow$:

If $\exists x_0$, s.t., $p(x_0)>0$, $q(x_0)=0$

$\Rightarrow p(x_0)\log\frac{p(x_0)}{q(x_0)} = +\infty$.

$\therefore KL(p||q) = \sum_{x\in X} p(x)\log\frac{p(x)}{q(x)} = +\infty$

here $0\log\frac{0}{q(x)} := 0$.

② $\Leftarrow$:

If $support(p)\subseteq support(q)$

$\Rightarrow q(x)>0$ when $p(x)>0$

$\Rightarrow p(x)\log\frac{p(x)}{q(x)}$ finite if $p(x)>0$

7.1 proof of 7.2:

① $n=2$: it holds. (by 7.4)

② if it holds for $n-1$.

③ then, $H(X_1,\cdots,X_{n-1},X_n)$

$= H(X_1,\cdots,X_{n-1}) + H(X_n|X_1,\cdots,X_{n-1})$

$= \sum_{i=1}^{n-1} H(X_i|X_1,\cdots,X_{i-1})$ (by ②) $+ H(X_n|X_1,\cdots,X_{n-1})$

$= \sum_{i=1}^{n} H(X_i|X_1,\cdots,X_{i-1})$ □

8. mutual information for discrete random variables

8.1 $I(X;Y) := KL(p(X,Y)\|p(X)p(Y))$

(p as a generic label for distri.)

$= KL(P_{X,Y}\|P_X P_Y) = KL(p(x,y)\|p(x)p(y))$, (p as fun)

8.3 $I(X;Y)\geq 0$, $I=0 \Leftrightarrow p(X,Y)=p(X)p(Y) \Leftrightarrow X,Y$ independent.

8.4 $I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$

proof: $I(X;Y) = \sum_{x,y} p(x,y)\log\frac{p(x|y)}{p(x)} = -H(X|Y) + H(X)$ □

8.5 ① $I(X;Y) = H(X,Y) - H(X|Y) - H(Y|X)$

② $I(X;Y) = H(X) + H(Y) - H(X,Y)$

9. $H(X) \geq H(X|Y)$ (by 8.3, 8.4)

② $H(X_n|X_1,\cdots,X_{n-1})$

$:= \sum_{x_1,\cdots,x_{n-1}} p(x_1,\cdots,x_{n-1}) H(P_{X_n|X_1=x_1,\cdots,X_{n-1}=x_{n-1}})$

$= \sum_{x_1,\cdots,x_{n-1}} p(x_1,\cdots,x_{n-1}) H(X_n|X_1=x_1,\cdots,X_{n-1}=x_{n-1})$ (by 7.5)

$= \sum_{x_1,\cdots,x_{n-1}} p(x_1,\cdots,x_{n-1})\left(-\sum_{x_n} p(x_n|x_1,\cdots,x_{n-1})\log p(x_n|x_1,\cdots,x_{n-1})\right)$

$= -\sum_{x_1,\cdots,x_n} p(x_1,\cdots,x_n)\log p(x_n|x_1,\cdots,x_{n-1})$

here $H(P_{X_n|X_1=x_1,\cdots,X_{n-1}=x_{n-1}})$

$= H(p(x_n|x_1,\cdots,x_{n-1}))$ ($p(x_n|x_1,\cdots,x_{n-1})$ as fun of x_n)

7.4 $H(X_1,X_2) = H(X_1) + H(X_2|X_1)$

proof: $H(X_1,X_2) = -\sum_{x_1,x_2} p(x_1,x_2)\log p(x_1,x_2)$

$= -\sum_{x_1,x_2} p(x_1,x_2)\log[p(x_1)p(x_2|x_1)]$

$= -\sum_{x_1,x_2} p(x_1,x_2)\log p(x_1)$

$-\sum_{x_1,x_2} p(x_1,x_2)\log p(x_2|x_1)$

(use 8.4, 7) $= H(X_1) + H(X_2|X_1)$

7.5 $H(X_1,\cdots,X_n) = H(X_1,\cdots,X_{n-1}) + H(X_n|X_1,\cdots,X_{n-1})$ 7.3

proof: left $= -\sum_{x_1,x_n} p(x_1,x_n)\log p(x_1,x_n)$

$= -\sum_{x_1,x_n} p(x_1,x_n)\log[p(x_1,x_{n-1})p(x_n|x_1,x_{n-1})]$ = right. □

6. The discrete distribution with maximum entropy is uniform distribution.

6.2 $H(p) \leq \log|X|$,

$H(p) = \log|X| \Leftrightarrow p(x)$ is uniform.

proof: ① $KL(p\|u) = \sum_x p(x)\log\frac{p(x)}{u(x)}$, $u(x) = \frac{1}{|X|}$

$= -H(p) - \sum p(x)\log u(x)$

$= -H(p) + \log|X|$

≥ 0

$\therefore H(p) \leq \log|X|$

② $H(p) = \log|X| \Leftrightarrow KL(p\|u) = 0 \Leftrightarrow p = u$ □

7. chain rule for entropy

7.2 $H(X_1,X_2,\cdots,X_n) = \sum_{i=1}^{n} H(X_i|X_1,\cdots,X_{i-1})$

$= H(X_1) + H(X_2|X_1) + \cdots + H(X_n|X_1,\cdots,X_{n-1})$

① $H(X_1,\cdots,X_n) := H(p(x_1,\cdots,x_n))$ ($p(x_1,\cdots,x_n)$ as function)

$= -\sum_{x_1,\cdots,x_n} p(x_1,\cdots,x_n)\log p(x_1,\cdots,x_n)$